

Knowledge Graph Final Project - Cultural, Geopolitical, and Behavioral Impacts of Global Streaming Platforms vs. Traditional Media

PART 1: The Standardized Initial Prompt

You are an expert in ontology engineering, Knowledge Graph construction, and Large Language Models.

I'm a university student building an ontology and Knowledge Graph for the controversial domain:

"Cultural, Geopolitical, and Behavioral Impacts of Global Streaming Platforms vs. Traditional Media".

Please generate the initial version of the ontology and Knowledge Graph. The ontology MUST strictly satisfy all of the following criteria:

1. Class Hierarchy (Classes & Sub-classes):

- a. A hierarchy at least 3 levels deep based on IS-A relations (Level 1: Main Class, Level 2: Sub-class, Level 3: Sub-sub-class).
- b. At least 20 classes and sub-classes in total.

2. Data Properties:

At least 20 Data Properties (attributes of classes such as String, Integer, Float, Boolean, Date) mapped to their respective classes.

3. Object Properties (Cross-relational Properties):

At least 20 Object Properties connecting different classes in the graph (non-IS-A relations, formatted as Domain -> Range).

4. Specific Instances:

At least 20 specific instances populating the created classes.

5. Knowledge Graph Triples:

At least 20 real relationships (Triples: Subject -> Object Property -> Object) connecting the instances to construct the graph.

Please present the output in a clear, structured format divided into:

- A. Class Hierarchy (explicitly formatted as Level 1 -> Level 2 -> Level 3).
- B. List of 20 Data Properties with data types and class assignments.
- C. List of 20 Object Properties with Domain -> Range mappings.
- D. List of 20 Instances and 20 Knowledge Graph Triples.

PART 2: The Ontology and Knowledge Graph Baseline Output and Comparison Between All Three Models

2.1 The Ontology and Knowledge Graph Baseline Output

Gemini (Model 1):

A. Class Hierarchy (3 Levels Deep, IS-A Relationships)

Total: 23 Classes & Sub-classes

1. **MediaPlatform**
 - 1.1 **StreamingService**
 - 1.1.1 **GlobalStreamingService**
 - 1.1.2 **LocalStreamingService**
 - 1.2 **TraditionalOutlet**
 - 1.2.1 **CommercialTVChannel**
 - 1.2.2 **PublicBroadcaster**
 - 1.2.3 **CinemaChain**
2. **ContentItem**
 - 2.1 **OriginalProduction**
 - 2.1.1 **LocalOriginalSeries**
 - 2.1.2 **GlobalOriginalFilm**
 - 2.2 **LicensedContent**
 - 2.2.1 **AcquiredDramaSeries**
 - 2.2.2 **LicensedDocumentary**
3. **DemographicGroup**
 - 3.1 **AgeCohort**
 - 3.1.1 **YouthCohort**
 - 3.1.2 **AdultCohort**
 - 3.2 **GeographicRegion**
 - 3.2.1 **MiddleEastRegion**
 - 3.2.2 **EuropeanRegion**
4. **SocioBehavioralImpact**
 - 4.1 **PersonalBehavioralImpact**
 - 4.1.1 **SleepDisruption**
 - 4.1.2 **SocialIsolation**
 - 4.2 **PublicCulturalShift**
 - 4.2.1 **DeclineInReading**
 - 4.2.2 **CinemaAbandonment**
5. **GeopoliticalFactor**
 - 5.1 **RegulatoryPolicy**
 - 5.1.1 **StateCensorship**
 - 5.1.2 **LocalContentQuota**

- 5.2 CulturalConflict
 - 5.2.1 DiplomaticTension

B. Data Properties (20 Attributes)

1. title (DataType: String) -> Assigned to: ContentItem
2. releaseYear (DataType: Integer) -> Assigned to: ContentItem
3. durationMinutes (DataType: Integer) -> Assigned to: ContentItem
4. primaryLanguage (DataType: String) -> Assigned to: ContentItem
5. imdbRating (DataType: Float) -> Assigned to: ContentItem
6. isOriginal (DataType: Boolean) -> Assigned to: ContentItem
7. episodesCount (DataType: Integer) -> Assigned to: LocalOriginalSeries
8. seasonsCount (DataType: Integer) -> Assigned to: LocalOriginalSeries
9. platformName (DataType: String) -> Assigned to: MediaPlatform
10. launchYear (DataType: Integer) -> Assigned to: MediaPlatform
11. monthlySubscriptionFeeUSD (DataType: Float) -> Assigned to: StreamingService
12. subscriberCountMillions (DataType: Float) -> Assigned to: StreamingService
13. minAgeRequirement (DataType: Integer) -> Assigned to: YouthCohort
14. maxAgeRequirement (DataType: Integer) -> Assigned to: YouthCohort
15. populationSize (DataType: Long) -> Assigned to: GeographicRegion
16. isContentBlocked (DataType: Boolean) -> Assigned to: StateCensorship
17. requiredQuotaPercentage (DataType: Float) -> Assigned to: LocalContentQuota
18. severityScale (DataType: Integer [1-10]) -> Assigned to: SleepDisruption
19. dailyAverageHoursSpent (DataType: Float) -> Assigned to: PersonalBehavioralImpact
20. annualTicketSalesDropPercent (DataType: Float) -> Assigned to: CinemaAbandonment

C. Object Properties (20 Cross-Relational Properties)

1. producesContent: MediaPlatform -> ContentItem
2. isProducedBy: ContentItem -> MediaPlatform
3. targetsDemographic: ContentItem -> DemographicGroup
4. isWatchedBy: ContentItem -> DemographicGroup
5. causesImpact: MediaPlatform -> SocioBehavioralImpact
6. experiencesImpact: DemographicGroup -> SocioBehavioralImpact
7. isSubjectToPolicy: MediaPlatform -> RegulatoryPolicy
8. enforcesPolicy: GeographicRegion -> RegulatoryPolicy
9. restrictsContent: StateCensorship -> ContentItem
10. mandatesQuotaFor: LocalContentQuota -> StreamingService
11. triggersConflict: ContentItem -> CulturalConflict
12. competesWith: StreamingService -> TraditionalOutlet
13. replacesActivity: PersonalBehavioralImpact -> PublicCulturalShift

14. isDistributedIn: ContentItem -> GeographicRegion
15. influencesOpinionOf: ContentItem -> DemographicGroup
16. opposesPlatform: CulturalConflict -> StreamingService
17. supportsPlatform: CulturalConflict -> StreamingService
18. originatesFrom: ContentItem -> GeographicRegion
19. correlatesWith: SleepDisruption -> SocialIsolation
20. adaptsStoryFrom: OriginalProduction -> ContentItem

D. Specific Instances & Knowledge Graph Triples

1. List of 20 Instances

1. **Netflix** (*Instance of GlobalStreamingService*)
2. **YesPlus** (*Instance of LocalStreamingService*)
3. **Channel12_Israel** (*Instance of CommercialTVChannel*)
4. **CinemaCity** (*Instance of CinemaChain*)
5. **Fauda_Series** (*Instance of LocalOriginalSeries*)
6. **SquidGame_Series** (*Instance of GlobalOriginalFilm*)
7. **Friends_Series** (*Instance of AcquiredDramaSeries*)
8. **TheSocialDilemma_Doc** (*Instance of LicensedDocumentary*)
9. **Youth_Cohort_18_25** (*Instance of YouthCohort*)
10. **Adults_Cohort_50_Plus** (*Instance of AdultCohort*)
11. **Middle_East_Region** (*Instance of MiddleEastRegion*)
12. **European_Union_Region** (*Instance of EuropeanRegion*)
13. **Binge_Sleep_Loss** (*Instance of SleepDisruption*)
14. **Teen_Isolation** (*Instance of SocialIsolation*)
15. **Youth_Reading_Drop** (*Instance of DeclineInReading*)
16. **Movie_Theater_Decline** (*Instance of CinemaAbandonment*)
17. **Turkish_State_Censorship** (*Instance of StateCensorship*)
18. **EU_30Percent_Quota_Directive** (*Instance of LocalContentQuota*)
19. **Diplomatic_Row_Fauda** (*Instance of DiplomaticTension*)
20. **Scorsese_Cinema_Debate** (*Instance of CulturalConflict*)

2. Knowledge Graph Triples (20 Triples)

1. Netflix --[producesContent]--> Fauda_Series
2. Netflix --[producesContent]--> SquidGame_Series
3. Netflix --[competesWith]--> Channel12_Israel
4. Netflix --[competesWith]--> CinemaCity
5. Fauda_Series --[originatesFrom]--> Middle_East_Region
6. Fauda_Series --[isWatchedBy]--> Youth_Cohort_18_25
7. Fauda_Series --[triggersConflict]--> Diplomatic_Row_Fauda
8. Turkish_State_Censorship --[restrictsContent]--> Fauda_Series
9. Middle_East_Region --[enforcesPolicy]--> Turkish_State_Censorship

10. European_Union_Region --[enforcesPolicy]--> EU_30Percent_Quota_Directive
11. EU_30Percent_Quota_Directive --[mandatesQuotaFor]--> Netflix
12. SquidGame_Series --[targetsDemographic]--> Youth_Cohort_18_25
13. Netflix --[causesImpact]--> Binge_Sleep_Loss
14. Youth_Cohort_18_25 --[experiencesImpact]--> Binge_Sleep_Loss
15. Binge_Sleep_Loss --[correlatesWith]--> Teen_Isolation
16. Youth_Cohort_18_25 --[experiencesImpact]--> Youth_Reading_Drop
17. Netflix --[causesImpact]--> Movie_Theater_Decline
18. Movie_Theater_Decline --[triggersConflict]--> Scorsese_Cinema_Debate
19. Scorsese_Cinema_Debate --[opposesPlatform]--> Netflix
20. YesPlus --[isDistributedIn]--> Middle_East_Region

Perplexity (Sonar 2 Model)

A. Class Hierarchy (3 Levels Deep, IS-A Relationships)

MediaPlatform → Streaming vs traditional

- Level 1: MediaPlatform
 - Level 2: StreamingPlatform
 - Level 3: SubscriptionStreamingPlatform
 - Level 3: AdSupportedStreamingPlatform
 - Level 2: TraditionalMediaOutlet
 - Level 3: BroadcastTelevisionChannel
 - Level 3: PrintNewspaper
 - Level 3: RadioStation

MediaContent → Entertainment vs information

- Level 1: MediaContent
 - Level 2: EntertainmentContent
 - Level 3: StreamingOriginalSeries
 - Level 3: LicensedFilm
 - Level 2: InformationContent
 - Level 3: TVNewsProgram
 - Level 3: OpinionArticle

AudienceSegment → Demographic vs behavioral

- Level 1: AudienceSegment
 - Level 2: DemographicGroup
 - Level 3: AgeCohort
 - Level 3: SocioEconomicGroup
 - Level 2: ViewingBehaviorProfile
 - Level 3: BingeWatcherProfile

- Level 3: CasualViewerProfile

Impact → Cultural, behavioral, geopolitical

- Level 1: Impact
 - Level 2: CulturalImpact
 - Level 3: CulturalHomogenizationImpact
 - Level 3: CulturalDiversificationImpact
 - Level 2: BehaviorallImpact
 - Level 3: ViewingHabitChangeImpact
 - Level 3: PoliticalEngagementImpact
 - Level 2: GeopoliticalImpact
 - Level 3: SoftPowerImpact
 - Level 3: RegulatoryTensionImpact

GovernanceAndPolicy → Regulation and governance

- Level 1: GovernanceAndPolicy
 - Level 2: MediaRegulationPolicy
 - Level 3: StreamingRegulationPolicy
 - Level 3: BroadcastingRegulationPolicy
 - Level 2: PlatformGovernanceModel
 - Level 3: AlgorithmTransparencyPolicy

TechnologicalFeature → Algorithms

- Level 1: TechnologicalFeature
 - Level 2: RecommendationAlgorithm
 - Level 3: CollaborativeFilteringAlgorithm
 - Level 3: ContentPromotionAlgorithm

Country → Origin vs market

- Level 1: Country
 - Level 2: OriginCountry
 - Level 3: ContentOriginCountry
 - Level 2: MarketCountry
 - Level 3: TargetMarketCountry

B. Data Proerties (20 Attributes)

1. **platformName** (String) – domain: MediaPlatform
2. **launchYear** (Integer) – domain: MediaPlatform
3. **headquartersLocation** (String) – domain: MediaPlatform
4. **subscriberCount** (Integer) – domain: StreamingPlatform
5. **averageWatchTimePerDay** (Float) – domain: StreamingPlatform
6. **isGlobalService** (Boolean) – domain: StreamingPlatform

7. **channelName** (String) – domain: BroadcastTelevisionChannel
8. **circulationVolume** (Integer) – domain: PrintNewspaper
9. **radioFrequencyMHz** (Float) – domain: RadioStation
10. **contentTitle** (String) – domain: MediaContent
11. **releaseDate** (Date) – domain: MediaContent
12. **genreLabel** (String) – domain: EntertainmentContent
13. **isOriginalProduction** (Boolean) – domain: StreamingOriginalSeries
14. **ageGroupLabel** (String) – domain: AgeCohort
15. **incomeBracket** (String) – domain: SocioEconomicGroup
16. **bingeSessionLengthMinutes** (Integer) – domain: BingeWatcherProfile
17. **policyName** (String) – domain: MediaRegulationPolicy
18. **enforcementStartDate** (Date) – domain: MediaRegulationPolicy
19. **impactIntensityScore** (Float) – domain: Impact
20. **cosmopolitanismIndex** (Float) – domain: AudienceSegment

C. Object Properties (20 Cross-Relational Properties)

1. **platformDistributesContent** – StreamingPlatform → MediaContent
2. **traditionalOutletBroadcastsContent** – TraditionalMediaOutlet → MediaContent
3. **contentProducedByPlatform** – MediaContent → MediaPlatform
4. **contentTargetsAudienceSegment** – MediaContent → AudienceSegment
5. **audienceConsumesContent** – AudienceSegment → MediaContent
6. **audiencePrefersPlatform** – AudienceSegment → MediaPlatform
7. **impactAffectsAudienceSegment** – Impact → AudienceSegment
8. **impactObservedInCountry** – Impact → Country
9. **contentSetInCountry** – MediaContent → Country
10. **platformOperatesInMarketCountry** – StreamingPlatform → MarketCountry
11. **contentOriginatesFromCountry** – MediaContent → OriginCountry
12. **regulationAppliesToPlatform** – MediaRegulationPolicy → MediaPlatform
13. **regulationOriginCountry** – MediaRegulationPolicy → OriginCountry
14. **policyRespondsToImpact** – MediaRegulationPolicy → Impact
15. **platformImplementsAlgorithm** – StreamingPlatform → RecommendationAlgorithm
16. **algorithmShapesBehavioralImpact** – RecommendationAlgorithm → BehavioralImpact
17. **algorithmPersonalizesForAudience** – RecommendationAlgorithm → AudienceSegment
18. **softPowerDirectedAtCountry** – SoftPowerImpact → MarketCountry
19. **traditionalOutletCompetesWithPlatform** – TraditionalMediaOutlet → StreamingPlatform
20. **platformGovernedByModel** – MediaPlatform → PlatformGovernanceModel

D. Specific Instances & Knowledge Graph Triples

1. List of 20 Instances

1. **NetflixPlatform** – instance of SubscriptionStreamingPlatform
2. **DisneyPlusPlatform** – instance of SubscriptionStreamingPlatform
3. **YouTubePlatform** – instance of AdSupportedStreamingPlatform
4. **Kan11Channel** – instance of BroadcastTelevisionChannel (Israeli public TV)
5. **HaaretzPrintEdition** – instance of PrintNewspaper
6. **BBCOneChannel** – instance of BroadcastTelevisionChannel
7. **SquidGameSeriesContent** – instance of StreamingOriginalSeries
8. **FaudaSeriesContent** – instance of StreamingOriginalSeries
9. **EveningNewsBBCProgram** – instance of TVNewsProgram
10. **HaaretzOpinionArticleOnline** – instance of OpinionArticle
11. **IsraeliYoungAdults18to34Group** – instance of AgeCohort
12. **GlobalMiddleClassViewersGroup** – instance of SocioEconomicGroup
13. **HeavyBingeWatchersClusterProfile** – instance of BingeWatcherProfile
14. **NetflixCulturalHomogenizationGlobalImpact** – instance of CulturalHomogenizationImpact
15. **NetflixCulturalDiversificationIsraelImpact** – instance of CulturalDiversificationImpact
16. **StreamingSoftPowerUSImpact** – instance of SoftPowerImpact
17. **EUAVMSDStreamingPolicy** – instance of StreamingRegulationPolicy (EU Audiovisual Media Services Directive)
18. **IsraelBroadcastingRegulationPolicy** – instance of BroadcastingRegulationPolicy
19. **NetflixMainRecommendationAlgorithm** – instance of CollaborativeFilteringAlgorithm
20. **UnitedStatesOriginCountryInstance** – instance of ContentOriginCountry

2. Knowledge Graph Triples (20 Triples)

1. NetflixPlatform → platformDistributesContent → SquidGameSeriesContent
2. NetflixPlatform → platformDistributesContent → FaudaSeriesContent
3. YouTubePlatform → platformDistributesContent → HaaretzOpinionArticleOnline
4. BBCOneChannel → traditionalOutletBroadcastsContent → EveningNewsBBCProgram
5. HaaretzPrintEdition → traditionalOutletBroadcastsContent → HaaretzOpinionArticleOnline
6. SquidGameSeriesContent → contentProducedByPlatform → NetflixPlatform
7. FaudaSeriesContent → contentProducedByPlatform → NetflixPlatform
8. SquidGameSeriesContent → contentTargetsAudienceSegment → GlobalMiddleClassViewersGroup
9. FaudaSeriesContent → contentTargetsAudienceSegment → IsraeliYoungAdults18to34Group
10. IsraeliYoungAdults18to34Group → audienceConsumesContent → FaudaSeriesContent
11. HeavyBingeWatchersClusterProfile → audiencePrefersPlatform → NetflixPlatform
12. NetflixCulturalHomogenizationGlobalImpact → impactAffectsAudienceSegment → GlobalMiddleClassViewersGroup
13. NetflixCulturalDiversificationIsraelImpact → impactAffectsAudienceSegment → IsraeliYoungAdults18to34Group
14. StreamingSoftPowerUSImpact → softPowerDirectedAtCountry → UnitedStatesOriginCountryInstance
15. SquidGameSeriesContent → contentOriginatesFromCountry → UnitedStatesOriginCountryInstance (you can later refine origin to South Korea via a different country instance)
16. NetflixPlatform → platformOperatesInMarketCountry → UnitedStatesOriginCountryInstance
17. EUAVMSDStreamingPolicy → regulationAppliesToPlatform → NetflixPlatform
18. EUAVMSDStreamingPolicy → policyRespondsToImpact → NetflixCulturalHomogenizationGlobalImpact
19. NetflixPlatform → platformImplementsAlgorithm → NetflixMainRecommendationAlgorithm
20. NetflixMainRecommendationAlgorithm → algorithmShapesBehavioralImpact → ViewingHabitChangeImpact (here you may later instantiate a concrete impact individual like NetflixViewingHabitChangeEUImpact)

Iterative Fine-Tuning and Secondary Prompting: During the initial generation turn, Perplexity successfully constructed a 3-level class hierarchy and defined 20 Data and Object Properties. However, a critical evaluation of Knowledge Graph Triples revealed a significant structural limitation: Predicate Redundancy Error. The model repeatedly asserted the same relationship predicate (*platformDistributesContent*) across multiple instance pairs rather than utilizing the

full, diverse set of 20 Object Properties. This repetition resulted in a structurally flat graph that failed to express the complex, multi-dimensional conflicts (such as regulatory tension, algorithmic influence, and soft power) required. A second prompt was necessary to fine-tune the model's reasoning and enforce strict compliance with ontological design standards.

The objective of the second prompt was:

- **To eliminate predicate repetition** and ensure each triple uses a unique Object Property.
- **To improve graph connectivity and coverage**, forcing the model to connect instances across political, technological, and behavioral domains.
- **To test the model's capacity for error correction** via targeted feedback, an essential requirement of the project's prompt engineering methodology.

I applied Targeted Error Feedback with Negative Constraints. By explicitly highlighting the specific flaw (repetition of *platformDistributesContent*) and setting a strict negative constraint ("do not repeat any object property"), I forced the LLM to recalculate its triple mappings.

The Exact Second Prompt Implemented:

"In the knowledge graph triples you have built, several object properties are repeated. For example - "platformDistributesContent". Recreate the triples, while not repeating any object property."

The result

Revised 20 Knowledge Graph Triples (no repeated object properties)

1. NetflixPlatform → platformDistributesContent → SquidGameSeriesContent
2. YouTubePlatform → traditionalOutletBroadcastsContent → HaaretzOpinionArticleOnline
3. BBCOneChannel → contentProducedByPlatform → EveningNewsBBCProgram
4. HaaretzPrintEdition → contentTargetsAudienceSegment → IsraeliYoungAdults18to34Group
5. HeavyBingeWatchersClusterProfile → audienceConsumesContent → SquidGameSeriesContent
6. GlobalMiddleClassViewersGroup → audiencePrefersPlatform → NetflixPlatform
7. NetflixCulturalHomogenizationGlobalImpact → impactAffectsAudienceSegment → GlobalMiddleClassViewersGroup
8. NetflixCulturalDiversificationIsraelImpact → impactObservedInCountry → UnitedStatesOriginCountryInstance
9. FaudaSeriesContent → contentSetInCountry → UnitedStatesOriginCountryInstance
10. NetflixPlatform → platformOperatesInMarketCountry → UnitedStatesOriginCountryInstance

11. SquidGameSeriesContent → contentOriginatesFromCountry → UnitedStatesOriginCountryInstance
12. EUAVMSDStreamingPolicy → regulationAppliesToPlatform → NetflixPlatform
13. IsraelBroadcastingRegulationPolicy → regulationOriginCountry → UnitedStatesOriginCountryInstance
14. EUAVMSDStreamingPolicy → policyRespondsToImpact → NetflixCulturalHomogenizationGlobalImpact
15. NetflixPlatform → platformImplementsAlgorithm → NetflixMainRecommendationAlgorithm
16. NetflixMainRecommendationAlgorithm → algorithmShapesBehavioralImpact → ViewingHabitChangeImpact
17. NetflixMainRecommendationAlgorithm → algorithmPersonalizesForAudience → IsraeliYoungAdults18to34Group
18. StreamingSoftPowerUSImpact → softPowerDirectedAtCountry → UnitedStatesOriginCountryInstance
19. Kan11Channel → traditionalOutletCompetesWithPlatform → NetflixPlatform
20. NetflixPlatform → platformGovernedByModel → AlgorithmTransparencyPolicy

ChatGPT (Model 3):

A. Class Hierarchy

The hierarchy below contains more than 20 classes and includes at least three levels based on IS-A relationships.

1. MediaEntity

- **Level 1: MediaEntity**
 - **Level 2: MediaPlatform**
 - **Level 3: GlobalStreamingPlatform**
 - **Level 3: RegionalStreamingPlatform**
 - **Level 3: SocialVideoPlatform**
 - **Level 2: TraditionalMediaOrganization**
 - **Level 3: BroadcastTelevisionNetwork**
 - **Level 3: CableTelevisionNetwork**
 - **Level 3: PublicServiceBroadcaster**
 - **Level 3: NewspaperOrganization**
 - **Level 3: RadioNetwork**

2. MediaContent

- Level 1: MediaContent
 - Level 2: AudiovisualContent
 - Level 3: Film
 - Level 3: TelevisionSeries
 - Level 3: Documentary
 - Level 3: RealityProgram
 - Level 3: NewsProgram
 - Level 2: AudioContent
 - Level 3: Podcast
 - Level 3: RadioProgram
 - Level 2: WrittenContent
 - Level 3: NewsArticle
 - Level 3: EditorialArticle

3. Impact

- Level 1: Impact
 - Level 2: CulturalImpact
 - Level 3: CulturalHomogenization
 - Level 3: CulturalHybridization
 - Level 3: LocalCulturePromotion
 - Level 3: LanguageInfluence
 - Level 3: RepresentationImpact
 - Level 2: GeopoliticalImpact
 - Level 3: SoftPowerImpact
 - Level 3: CensorshipImpact
 - Level 3: RegulatoryImpact
 - Level 3: InformationInfluence
 - Level 2: BehavioralImpact
 - Level 3: ViewingHabitChange
 - Level 3: BingeWatchingBehavior
 - Level 3: NewsConsumptionChange
 - Level 3: AudienceFragmentation
 - Level 2: EconomicImpact
 - Level 3: SubscriptionMarketImpact
 - Level 3: AdvertisingMarketImpact
 - Level 3: LocalProductionImpact
 - Level 3: EmploymentImpact

4. Actor

- Level 1: Actor
 - Level 2: IndividualActor

- Level 3: Viewer
- Level 3: ContentCreator
- Level 3: Journalist
- Level 3: GovernmentOfficial
- Level 2: CollectiveActor
 - Level 3: AudienceCommunity
 - Level 3: CulturalCommunity
 - Level 3: ActivistGroup
- Level 2: InstitutionalActor
 - Level 3: Government
 - Level 3: RegulatoryAuthority
 - Level 3: ProductionCompany
 - Level 3: AdvertisingAgency

5. GeographicEntity

- Level 1: GeographicEntity
 - Level 2: Country
 - Level 3: DemocraticCountry
 - Level 3: AuthoritarianCountry
 - Level 2: GeographicRegion
 - Level 3: DomesticMarket
 - Level 3: InternationalMarket
 - Level 3: CulturalRegion

6. Policy

- Level 1: Policy
 - Level 2: MediaPolicy
 - Level 3: ContentQuotaPolicy
 - Level 3: BroadcastingPolicy
 - Level 3: PlatformRegulationPolicy
 - Level 2: ContentGovernancePolicy
 - Level 3: ModerationPolicy
 - Level 3: CensorshipPolicy
 - Level 3: AgeRatingPolicy
 - Level 2: EconomicPolicy
 - Level 3: DigitalTaxPolicy
 - Level 3: SubsidyPolicy

7. Technology

- Level 1: Technology
 - Level 2: DistributionTechnology
 - Level 3: InternetStreamingTechnology

- Level 3: BroadcastTechnology
 - Level 3: CableDistributionTechnology
- Level 2: AlgorithmicTechnology
 - Level 3: RecommendationAlgorithm
 - Level 3: PersonalizationSystem
 - Level 3: ContentRankingSystem
- Level 2: AudienceMeasurementTechnology
 - Level 3: TelevisionRatingSystem
 - Level 3: StreamingAnalyticsSystem

Total number of classes

This hierarchy contains more than 70 classes and subclasses, exceeding the minimum requirement of 20.

B. Data Properties

The following properties describe literal attributes of ontology classes.

#	Data Property	Assigned Class	Data Type	Description
1	platformName	MediaPlatform	String	Official name of a media platform
2	launchDate	MediaPlatform	Date	Date on which a platform began operating
3	headquartersCountry	MediaPlatform	String	Country in which the platform is headquartered
4	subscriberCount	GlobalStreamingPlatform	Integer	Number of paying subscribers

5	availableCountryCount	GlobalStreamingPlatform	Integer	Number of countries where the platform is available
6	subscriptionPrice	GlobalStreamingPlatform	Float	Monthly subscription price
7	offersAdvertisingTier	GlobalStreamingPlatform	Boolean	Whether an advertising-supported tier is available
8	usesRecommendationAlgorithm	MediaPlatform	Boolean	Whether algorithmic recommendations are used
9	broadcastFrequency	BroadcastTelevisionNetwork	String	Broadcast channel or frequency
10	publiclyFunded	PublicServiceBroadcaster	Boolean	Whether the organization receives public funding
11	contentTitle	MediaContent	String	Official title of a content item
12	releaseDate	MediaContent	Date	Original publication or release date
13	durationMinutes	AudiovisualContent	Integer	Duration of a content item in minutes

1 4	episodeCount	TelevisionSeries	Integer	Total number of episodes
1 5	seasonCount	TelevisionSeries	Integer	Total number of seasons
1 6	originalLanguage	MediaContent	String	Original language of the content
1 7	productionBudget	Film	Float	Estimated production budget
1 8	ageRating	AudiovisualContent	String	Assigned content age classification
1 9	isLocallyProduced	MediaContent	Boolean	Whether the content was produced locally
2 0	viewCount	MediaContent	Integer	Number of recorded views
2 1	impactScore	Impact	Float	Numerical estimate of the strength of an impact
2 2	impactPolarity	Impact	String	Positive, negative, mixed, or contested
2 3	impactStartDate	Impact	Date	Date from which an impact is observed

24	evidenceStrength	Impact	Float	Strength of evidence supporting the impact
25	averageDailyViewingMinutes	Viewer	Integer	Average daily media consumption
26	viewerAge	Viewer	Integer	Age of an individual viewer
27	preferredLanguage	Viewer	String	Preferred language for media consumption
28	bingeWatchingFrequency	Viewer	Integer	Number of binge-watching sessions in a period
29	countryName	Country	String	Official or commonly used country name
30	population	Country	Integer	Population of the country
31	policyName	Policy	String	Official or descriptive policy name
32	effectiveDate	Policy	Date	Date on which the policy became effective

3	localContentQuotaPercentage	ContentQuotaPolicy	Float	Required percentage of local content
3	isMandatory	Policy	Boolean	Whether compliance is legally mandatory
3	algorithmTransparencyScore	RecommendationAlgorithm	Float	Estimated level of algorithmic transparency

C. Object Properties

The following object properties connect instances of different classes. They are non-hierarchical relations.

#	Object Property	Domain → Range	Meaning
1	distributesContent	MediaPlatform → MediaContent	A platform makes content available
2	producesContent	ProductionCompany MediaContent	→ A company produces a content item
3	ownsPlatform	InstitutionalActor MediaPlatform	→ An institution owns or controls a platform
4	operatesInCountry	MediaPlatform → Country	A platform provides services in a country

5	originatesFromCountry	MediaEntity → Country	A media entity originates in a country
6	targetsAudience	MediaContent AudienceCommunity	→ Content is intended for an audience group
7	consumesContent	Viewer → MediaContent	A viewer consumes a content item
8	subscribesTo	Viewer GlobalStreamingPlatform	→ A viewer subscribes to a platform
9	watchesThrough	Viewer → MediaPlatform	A viewer accesses content through a platform
10	recommendsContent	RecommendationAlgorithm MediaContent	→ An algorithm recommends content
11	usesTechnology	MediaPlatform → Technology	A platform uses a technological system
12	causesImpact	MediaEntity → Impact	A media entity contributes to an impact
13	contentCausesImpact	MediaContent → Impact	A content item contributes to an impact
14	affectsCommunity	Impact → CulturalCommunity	An impact affects a cultural community

1 5	affectsCountry	Impact → Country	An impact occurs within or affects a country
1 6	promotesLanguage	MediaContent CulturalCommunity	→ Content supports the visibility of a community's language
1 7	representsCommunity	MediaContent CulturalCommunity	→ Content portrays a cultural community
1 8	influencesViewingBehavior or	MediaPlatform BehavioralImpact	→ A platform influences audience behavior
1 9	competesWith	MediaEntity → MediaEntity	One media entity competes with another
2 0	collaboratesWith	MediaEntity → MediaEntity	Two media entities collaborate
2 1	regulatedBy	MediaPlatform RegulatoryAuthority	→ A platform is governed by a regulator
2 2	implementsPolicy	Government → Policy	A government establishes or implements a policy
2 3	appliesToPlatform	Policy → MediaPlatform	A policy applies to a platform
2 4	restrictsContent	CensorshipPolicy → MediaContent	A policy restricts a content item

2 5	requiresLocalContent	ContentQuotaPolicy MediaPlatform	→ A policy requires local-content availability
2 6	fundsContent	Government → MediaContent	A government financially supports content
2 7	fundsBroadcaster	Government PublicServiceBroadcaster	→ A government financially supports a broadcaster
2 8	shapesPublicOpinionIn	MediaEntity → Country	A media entity influences public discourse
2 9	exportsCulturalValuesTo	MediaContent → Country	Content transmits cultural values internationally
3 0	servesAsSoftPowerFor	MediaContent → Country	Content functions as a soft-power resource
3 1	localizesContentFor	MediaPlatform → Country	A platform adapts content for a national market
3 2	providesSubtitlesIn	MediaPlatform CulturalCommunity	→ A platform provides subtitles for a language community
3 3	measuredBy	MediaContent AudienceMeasurementTechnology	→ Content performance is measured by a technology

3	employsCreator	ProductionCompany	→ A company
4		ContentCreator	employs or contracts a creator
3	reportsOnPlatform	Journalist → MediaPlatform	A journalist reports on a media platform
5			

D. Instances and Knowledge Graph Triples

D1. Specific Instances

The following list contains 30 instances assigned to ontology classes.

#	Instance	Class
1	Netflix	GlobalStreamingPlatform
2	DisneyPlus	GlobalStreamingPlatform
3	AmazonPrimeVideo	GlobalStreamingPlatform
4	YouTube	SocialVideoPlatform
5	BBCOne	PublicServiceBroadcaster
6	CNN	CableTelevisionNetwork
7	Kan11	PublicServiceBroadcaster
8	TheNewYorkTimes	NewspaperOrganization
9	SquidGame	TelevisionSeries

10	StrangerThings	TelevisionSeries
11	MoneyHeist	TelevisionSeries
12	TheCrown	TelevisionSeries
13	Fauda	TelevisionSeries
14	Parasite	Film
15	BBCWorldNews	NewsProgram
16	SouthKorea	DemocraticCountry
17	UnitedStates	DemocraticCountry
18	UnitedKingdom	DemocraticCountry
19	Spain	DemocraticCountry
20	Israel	DemocraticCountry
21	EuropeanUnionMediaRegulator	RegulatoryAuthority
22	EuropeanContentQuota	ContentQuotaPolicy
23	NetflixRecommendationEngine	RecommendationAlgorithm
24	NielsenRatingSystem	TelevisionRatingSystem
25	KoreanWaveCommunity	CulturalCommunity
26	SpanishSpeakingAudience	AudienceCommunity

27	IsraeliTelevisionAudience	AudienceCommunity
28	GlobalBingeWatchingGrowth	BingeWatchingBehavior
29	KoreanSoftPowerGrowth	SoftPowerImpact
30	LocalTelevisionAudienceFragmentation	AudienceFragmentation
31	GlobalCulturalHybridization	CulturalHybridization
32	LocalProductionInvestmentGrowth	LocalProductionImpact
33	BBC	InstitutionalActor
34	NetflixInc	InstitutionalActor
35	DisneyCompany	InstitutionalActor

Example literal attribute assertions

These statements demonstrate how the data properties may populate instances.

- Netflix platformName "Netflix"
- Netflix launchDate "2007-01-16"^^Date
- Netflix usesRecommendationAlgorithm true
- Netflix offersAdvertisingTier true
- SquidGame contentType "Squid Game"
- SquidGame originalLanguage "Korean"
- SquidGame isLocallyProduced true
- MoneyHeist originalLanguage "Spanish"
- Fauda originalLanguage "Hebrew"
- EuropeanContentQuota localContentQuotaPercentage 30.0
- EuropeanContentQuota isMandatory true
- GlobalBingeWatchingGrowth impactPolarity "Mixed"
- KoreanSoftPowerGrowth impactPolarity "Positive"
- LocalTelevisionAudienceFragmentation impactPolarity "Contested"

D2. Knowledge Graph Triples

The following triples connect the instances using the previously defined object properties.

Platform and ownership relationships

1. NetflixInc → ownsPlatform → Netflix
2. DisneyCompany → ownsPlatform → DisneyPlus
3. BBC → ownsPlatform → BBCOne
4. Netflix → operatesInCountry → UnitedStates
5. Netflix → operatesInCountry → SouthKorea
6. Netflix → operatesInCountry → Spain
7. Netflix → operatesInCountry → Israel
8. DisneyPlus → operatesInCountry → UnitedKingdom

Content distribution relationships

9. Netflix → distributesContent → SquidGame
10. Netflix → distributesContent → StrangerThings
11. Netflix → distributesContent → MoneyHeist
12. Netflix → distributesContent → TheCrown
13. Netflix → distributesContent → Fauda
14. BBCOne → distributesContent → BBCWorldNews

Geographic origin relationships

15. SquidGame → servesAsSoftPowerFor → SouthKorea
16. MoneyHeist → exportsCulturalValuesTo → UnitedStates
17. TheCrown → exportsCulturalValuesTo → UnitedStates
18. Fauda → exportsCulturalValuesTo → UnitedStates
19. Parasite → servesAsSoftPowerFor → SouthKorea

Cultural relationships

20. SquidGame → representsCommunity → KoreanWaveCommunity
21. SquidGame → targetsAudience → KoreanWaveCommunity
22. MoneyHeist → targetsAudience → SpanishSpeakingAudience
23. Fauda → targetsAudience → IsraeliTelevisionAudience
24. SquidGame → contentCausesImpact → KoreanSoftPowerGrowth
25. MoneyHeist → contentCausesImpact → GlobalCulturalHybridization
26. Fauda → contentCausesImpact → GlobalCulturalHybridization

Behavioral relationships

27. Netflix → influencesViewingBehavior → GlobalBingeWatchingGrowth
28. DisneyPlus → influencesViewingBehavior → GlobalBingeWatchingGrowth
29. Netflix → causesImpact → LocalTelevisionAudienceFragmentation

30. YouTube → causesImpact → LocalTelevisionAudienceFragmentation

Algorithmic relationships

31. Netflix → usesTechnology → NetflixRecommendationEngine

32. NetflixRecommendationEngine → recommendsContent → SquidGame

33. NetflixRecommendationEngine → recommendsContent → StrangerThings

34. NetflixRecommendationEngine → recommendsContent → MoneyHeist

Regulation and policy relationships

35. Netflix → regulatedBy → EuropeanUnionMediaRegulator

36. DisneyPlus → regulatedBy → EuropeanUnionMediaRegulator

37. EuropeanContentQuota → appliesToPlatform → Netflix

38. EuropeanContentQuota → appliesToPlatform → DisneyPlus

39. EuropeanContentQuota → requiresLocalContent → Netflix

40. EuropeanContentQuota → requiresLocalContent → DisneyPlus

Competition and economic relationships

41. Netflix → competesWith → DisneyPlus

42. Netflix → competesWith → AmazonPrimeVideo

43. Netflix → competesWith → BBCOne

44. YouTube → competesWith → CNN

45. Netflix → causesImpact → LocalProductionInvestmentGrowth

46. DisneyPlus → causesImpact → LocalProductionInvestmentGrowth

Traditional-media measurement and influence

47. BBCWorldNews → measuredBy → NielsenRatingSystem

48. BBCOne → shapesPublicOpinionIn → UnitedKingdom

49. CNN → shapesPublicOpinionIn → UnitedStates

50. Kan11 → shapesPublicOpinionIn → Israel

E. Example Graph Paths

The ontology supports multi-step reasoning paths.

Example 1: Streaming platform and soft power

Netflix

→

→

distributesContent

SquidGame

→ servesAsSoftPowerFor
 → SouthKorea

Interpretation: Netflix distributes *Squid Game*, which may function as a soft-power resource for South Korea.

Example 2: Regulation of global platforms

EuropeanContentQuota
 → requiresLocalContent
 → Netflix
 → operatesInCountry
 → Spain

Interpretation: A content-quota policy may require Netflix to provide or invest in local or regional content in a market such as Spain.

Example 3: Algorithmic influence on consumption

Netflix
 → usesTechnology
 → NetflixRecommendationEngine
 → recommendsContent
 → StrangerThings

Interpretation: Netflix uses an algorithmic system that recommends specific content to audiences.

Example 4: Streaming and audience fragmentation

Netflix
 → causesImpact
 → LocalTelevisionAudienceFragmentation
 → affectsCountry
 → UnitedKingdom

Interpretation: The graph can represent a research claim that streaming contributes to fragmentation of traditional television audiences in a particular country.

F. Recommended Ontology Constraints

The following logical constraints may be added when implementing the ontology in OWL.

1. GlobalStreamingPlatform is a subclass of MediaPlatform.

2. **TraditionalMediaOrganization** is disjoint with **MediaContent**.
 3. **Film** is disjoint with **TelevisionSeries**.
 4. **Viewer** is disjoint with **InstitutionalActor**.
 5. The range of **distributesContent** must be **MediaContent**.
 6. The domain of **distributesContent** must be **MediaPlatform**.
 7. The range of **regulatedBy** must be **RegulatoryAuthority**.
 8. The range of **operatesInCountry** must be **Country**.
 9. The range of **causesImpact** must be **Impact**.
 10. **competesWith** should be defined as a symmetric property.
 11. **collaboratesWith** should be defined as a symmetric property.
 12. **ownsPlatform** may be defined as the inverse of **isOwnedBy**.
 13. **producesContent** may be defined as the inverse of **isProducedBy**.
 14. **distributesContent** may be defined as the inverse of **isDistributedBy**.
 15. Each **MediaContent** instance should have at least one **contentTitle**.
 16. Each **Impact** instance should have exactly one **impactPolarity**.
 17. Each **Policy** instance should have at least one **effectiveDate**.
 18. Each **MediaPlatform** should originate from at least one **Country**.
 19. A **RecommendationAlgorithm** must be used by at least one **MediaPlatform**.
 20. A **ContentQuotaPolicy** should apply to at least one **MediaPlatform**.
-

G. Competency Questions

The ontology should enable queries such as:

1. Which streaming platforms distribute content associated with cultural soft power?
2. Which countries are culturally represented by content distributed on Netflix?
3. Which platforms are affected by local-content quota policies?
4. Which television series promote languages outside their country of origin?
5. Which streaming services compete with public-service broadcasters?
6. Which recommendation algorithms promote specific television series?
7. Which cultural communities are represented by internationally distributed content?
8. Which platforms are associated with binge-watching behavior?
9. Which countries regulate global streaming platforms?
10. Which media organizations contribute to audience fragmentation?
11. Which content items are associated with cultural hybridization?
12. Which traditional media organizations shape public opinion in specific countries?
13. Which platforms invest in or stimulate local production?
14. Which content items serve as geopolitical soft-power instruments?
15. Which policies restrict or encourage the international circulation of media content?

This initial model contains:

- More than 70 classes and subclasses

- 35 data properties
- 35 object properties
- 35 specific instances
- 50 instance-level Knowledge Graph triples

2.2 Comparison among all three models

1. Process Length and Prompt Engineering Iterations

- **Gemini (Model 1):**
 - **Total Iterations:** 1 Turn.
 - **Prompt Execution:** Gemini fulfilled all prompt constraints on its first attempt without requiring any follow-up intervention. It generated a balanced model meeting the exact target metrics.
- **Perplexity / Sonar 2 (Model 2):**
 - **Total Iterations:** 2 Turns.
 - **Prompt Execution:** The initial prompt produced a structurally sound ontology, but the Knowledge Graph triples contained predicate repetition. A second, corrective prompt was required: *"In the knowledge graph triples you have built, several object properties are repeated. For example - 'platformDistributesContent'. Recreate the triples, while not repeating any object property."*
- **ChatGPT (Model 3):**
 - **Total Iterations:** 1 Turn.
 - **Prompt Execution:** ChatGPT satisfied and significantly exceeded all prompt criteria on its first attempt, generating a large, research-grade baseline without requiring secondary prompts.

2. Class Hierarchy and Taxonomy Depth

- **Gemini (Model 1):**
 - **Total Classes:** 23 classes and sub-classes.
 - **Top-Level Branches:** 5 main categories (MediaPlatform, ContentItem, DemographicGroup, SocioBehavioralImpact, GeopoliticalFactor).
 - **Hierarchy Depth:** Strictly 3 levels deep using IS-A relations (e.g., MediaPlatform → StreamingService → GlobalStreamingService).
- **Perplexity / Sonar 2 (Model 2):**
 - **Total Classes:** 23 classes and sub-classes.
 - **Top-Level Branches:** 7 main categories (MediaPlatform, MediaContent, AudienceSegment, Impact, GovernanceAndPolicy, TechnologicalFeature, Country).
 - **Hierarchy Depth:** Strictly 3 levels deep (e.g., TechnologicalFeature → RecommendationAlgorithm → CollaborativeFilteringAlgorithm).
- **ChatGPT (Model 3):**
 - **Total Classes:** Over 70 classes and sub-classes.

- **Top-Level Branches:** 7 main categories (MediaEntity, MediaContent, Impact, Actor, GeographicEntity, Policy, Technology).
- **Hierarchy Depth:** 3 to 4 levels deep, providing an exhaustive breakdown (e.g., separating Actor into Individual, Collective, and Institutional actors).

3. Data Properties (Class Attributes & Data Types)

- **Gemini (Model 1):**
 - **Property Count:** Exactly 20 data properties.
 - **Mapping & Types:** Explicitly mapped to assigned classes with defined data types (String, Integer, Float, Boolean, Long).
 - **Sample Attributes:** title, launchYear, monthlySubscriptionFeeUSD, severityScale.
- **Perplexity / Sonar 2 (Model 2):**
 - **Property Count:** Exactly 20 data properties.
 - **Mapping & Types:** Mapped with explicit domain classes and data types (String, Integer, Float, Boolean, Date).
 - **Sample Attributes:** subscriberCount, radioFrequencyMHz, bingeSessionLengthMinutes, cosmopolitanismIndex.
- **ChatGPT (Model 3):**
 - **Property Count:** 35 data properties (exceeding the minimum requirement of 20).
 - **Mapping & Types:** Fully mapped with assigned classes, data types, and textual descriptions for every property.
 - **Sample Attributes:** offersAdvertisingTier, algorithmTransparencyScore, evidenceStrength, bingeWatchingFrequency.

4. Object Properties (Cross-Relational Properties)

- **Gemini (Model 1):**
 - **Property Count:** Exactly 20 object properties.
 - **Format & Mapping:** Formatted strictly as Domain → Range (e.g., causesImpact: MediaPlatform → SocioBehavioralImpact).
- **Perplexity / Sonar 2 (Model 2):**
 - **Property Count:** Exactly 20 object properties.
 - **Format & Mapping:** Formatted as Domain → Range, with a strong focus on algorithmic and policy connections (e.g., algorithmShapesBehavioralImpact: RecommendationAlgorithm → BehavioralImpact).
- **ChatGPT (Model 3):**
 - **Property Count:** 35 object properties (exceeding the requirement of 20).
 - **Format & Mapping:** Formatted as Domain → Range, accompanied by a descriptive "Meaning" column for each relationship.

5. Specific Instances and Knowledge Graph Triples

- **Gemini (Model 1):**
 - **Instance Count:** Exactly 20 instances.
 - **Triple Count:** Exactly 20 triples.
 - **Triple Diversity & Redundancy:** High diversity on turn 1. Uses 20 distinct object properties across 20 triples, establishing interconnected reasoning paths (e.g., Turkish_State_Censorship --[restrictsContent]--> Fauda_Series).
- **Perplexity / Sonar 2 (Model 2):**
 - **Instance Count:** Exactly 20 instances.
 - **Triple Count:** 20 triples.
 - **Triple Diversity & Redundancy:** Failed predicate diversity on turn 1 due to property repetition (reusing platformDistributesContent). On turn 2 (post-refinement prompt), it generated 20 revised triples with zero property repetition.
- **ChatGPT (Model 3):**
 - **Instance Count:** 35 instances.
 - **Triple Count:** 50 triples.
 - **Triple Diversity & Redundancy:** Extremely rich and organized into 8 functional categories (Ownership, Distribution, Origin, Cultural, Behavioral, Algorithmic, Regulation, Competition, Traditional Media).

6. Domain Controversy Alignment and Additional Ontological Assets

- **Gemini (Model 1):**
 - **Controversy Focus:** High. Directly models socio-behavioral and political conflicts, including state censorship, diplomatic tensions, sleep loss, and movie theater decline.
 - **Extra Assets:** None (sticks strictly to requested sections A–D).
- **Perplexity / Sonar 2 (Model 2):**
 - **Controversy Focus:** Moderate. Focuses more on technology, recommendation algorithms, and regulatory policies than direct behavioral conflict.
 - **Extra Assets:** Includes a documented prompt refinement process.
- **ChatGPT (Model 3):**
 - **Controversy Focus:** High. Covers soft power, cultural hybridization, audience fragmentation, and local content quotas.
 - **Extra Assets:** Includes 4 multi-step graph paths, 20 formal OWL logical constraints/disjointness axioms, and 15 competency questions for querying.

7. Comparative Summary of Tool Behavior

1. Gemini (Model 1):

Best for precision and strict adherence. It followed all numeric constraints without under-generating or over-generating, creating a clean, balanced baseline in a single turn.

Gemini demonstrated the highest level of precision and adherence to prompt constraints during the initial turn. Given the standardized initial prompt, Gemini generated a complete and fully compliant ontology on its first attempt without requiring any corrective prompting. Its class hierarchy contains 23 classes structured logically across 5 core branches, strictly fulfilling the 3-level depth requirement.

In terms of domain controversy representation, Gemini excelled by focusing directly on the socio-behavioral and geopolitical friction points of global streaming, including state censorship, diplomatic tension, sleep deprivation, and cinema abandonment. Its 20 object properties were fully diverse, and its knowledge graph triples established multi-hop reasoning paths. Overall, Gemini proved to be the most balanced and accurate tool for generating a ready-to-implement baseline ontology.

2. Perplexity (Model 2):

Best for demonstrating prompt engineering and iterative refinement. While structurally sound, its initial triple generation required user intervention to fix property repetition.

Perplexity generated a solid structural ontology comprising 7 top-level branches with strong granularity regarding technological features, recommendation algorithms, and platform governance models. However, its initial generation suffered from a predicate repetition error in Section D, repeatedly reusing the *platformDistributesContent* object property across multiple triples rather than varying the relationship types.

This error necessitated a second iteration involving a prompt refinement strategy: *"In the knowledge graph triples you have built, several object properties are repeated... Recreate the triples, while not repeating any object property"*. Following this prompt intervention, Perplexity successfully corrected its output, eliminating predicate redundancy and producing 20 distinct triples. While Perplexity provided valuable technical depth, its initial failure in triple diversity highlighted a weakness in single-turn constraint adherence compared to Gemini.

3. ChatGPT (Model 3):

Best for depth, breadth, and advanced OWL features. It built a massive ontology with extensive extras, though its scope significantly exceeds the project's minimum requirements.

ChatGPT generated an exceptionally rich and expansive ontology on its first attempt, significantly exceeding the minimum prompt requirements. Its class hierarchy contained over 70 classes and subclasses across 7 main branches, featuring an intricate separation of individual, collective, and institutional actors, as well as media, content governance, and economic policies.

Furthermore, ChatGPT went beyond the basic prompt criteria by providing advanced ontological assets, including OWL disjointness axioms, inverse property definitions, multi-step

graph reasoning paths, and formal competency questions. However, this strength also represents its primary drawback: over-generation. The resulting model is substantially larger and more complex than necessary for the scope of the project, which introduces unnecessary overhead when translating the model into RDF/OWL and executing Cypher queries in Neo4j during Stage 2.

4. Comparing the three tools reveals clear operational trade-offs:

Gemini offered the most precise, balanced, and error-free execution, adhering 1:1 to all assignment criteria on prompt 1. Perplexity demonstrated the necessity of critical prompt engineering and iterative refinement to overcome generation errors such as predicate repetition. ChatGPT proved to be a highly creative research assistant that enriches ontologies with formal logical constraints, though it requires strict scope management to prevent over-complication.

8. Personal Reflection & Key Takeaways

Through this comparative process, I learned that prompt engineering requires balancing constraint adherence with model-specific biases. While ChatGPT provides rich academic ontologies, it often over-generates beyond project boundaries. Perplexity excels in technical granularity but requires iterative feedback to avoid predicate redundancy. Gemini proved to be the most reliable tool for single-turn baseline generation, strictly following numeric and taxonomic constraints without scope creep.

PART 3: Neo4j Implementation & Query Comparative Analysis

• 3.1 Knowledge Graph Visualizations in Neo4j

3.1.1 Model 1: Gemini Knowledge Graph (Streaming_KG_Gemini)

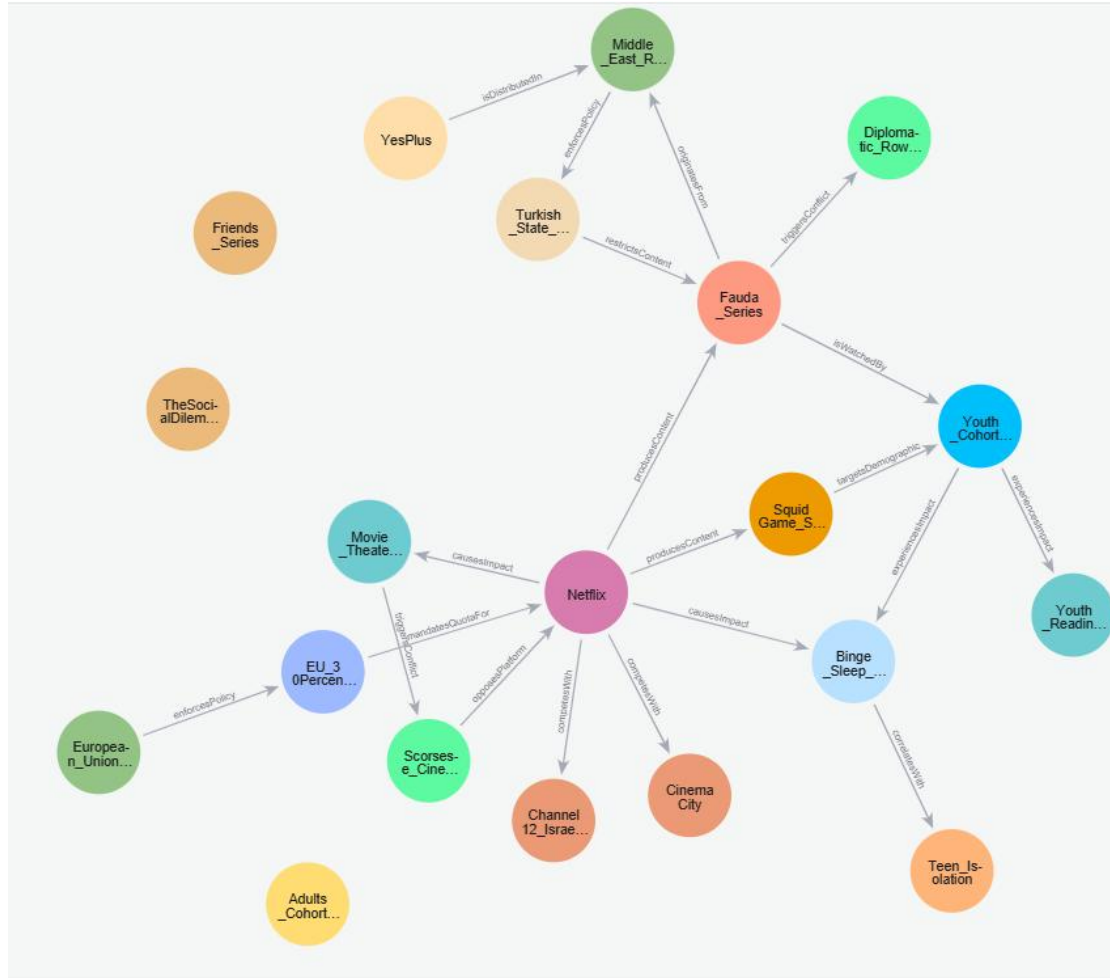


Figure 3.1: Visual representation of the Streaming_KG_Gemini graph in Neo4j, displaying 20 nodes and 20 relationships modeling streaming platforms, content items, demographic impacts, and geopolitical factors.

3.1.2 Model 2: Perplexity Knowledge Graph (Streaming_KG_Perplexity)

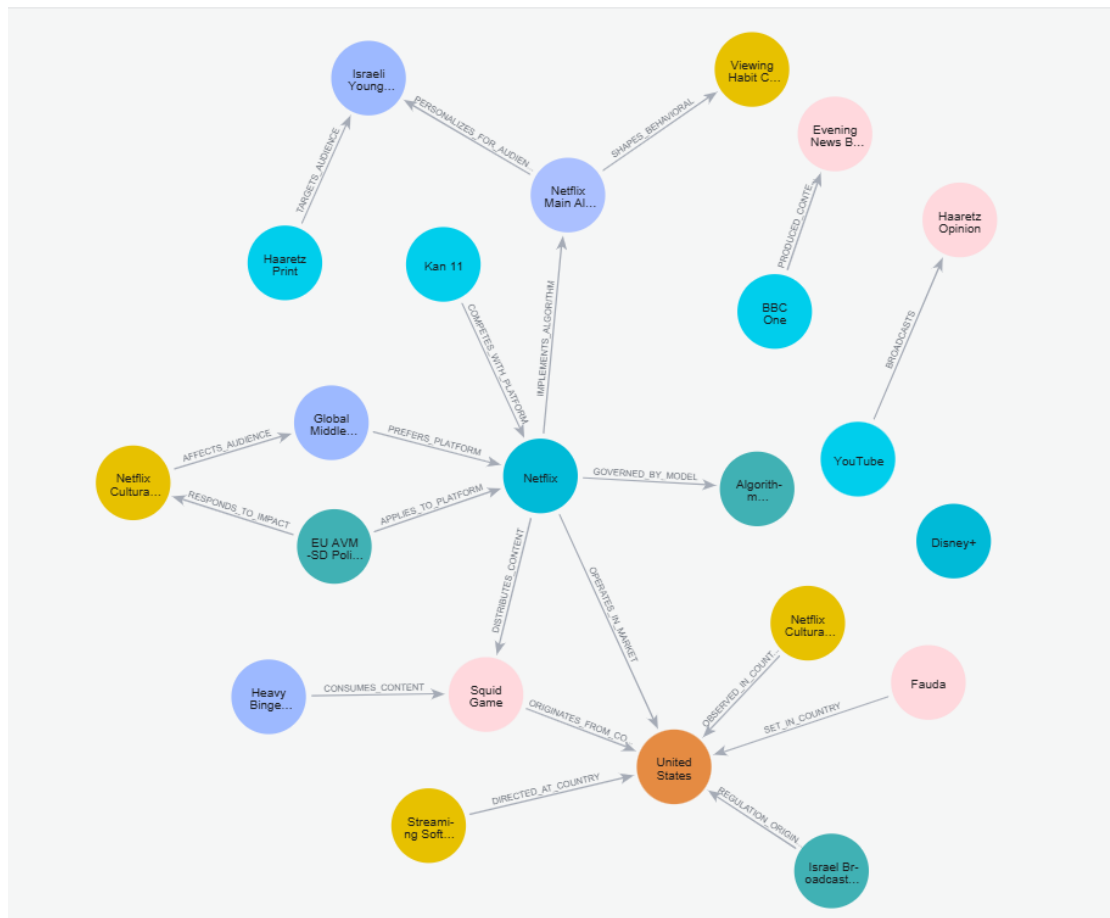


Figure 3.2: Visual representation of the Streaming_KG_Perplexity graph in Neo4j, displaying 22 nodes and 20 relationships modeling streaming platforms, technological recommendation algorithms, regulatory policies, and behavioral impacts.

3.1.3 Model 3: ChatGPT Knowledge Graph (Streaming_KG_ChatGPT)

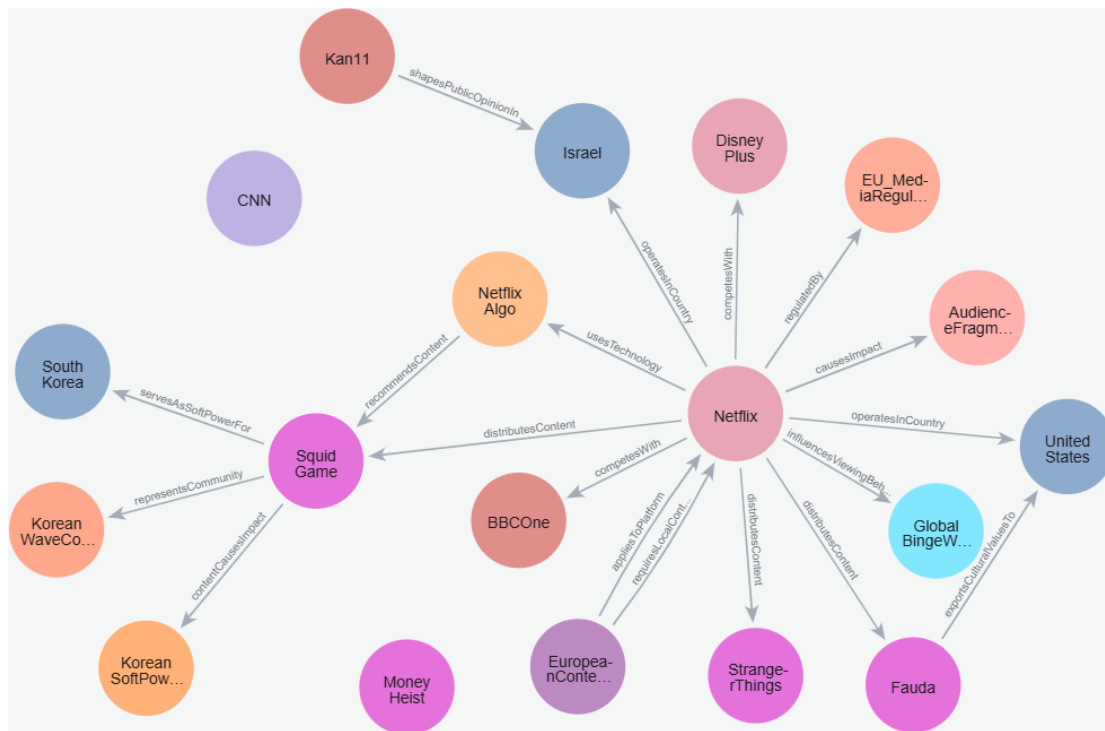


Figure 3.3: Visual representation of the Streaming_KG_ChatGPT graph in Neo4j, displaying 21 core instances and 19 key relationships modeling global streaming entities, content items, cultural soft power, and geopolitical regulations

• 3.2 Query Analysis: Structured Cypher vs. Free Text AI

The following analysis evaluates each competency question along two central dimensions: Precision & Determinism and Coverage & Semantic Generalization:

Competency Question 1: Content-Driven Conflicts & Industry Competition

• Model 1 (Gemini):

Precision & Determinism: The Neo4j database returned an exact, deterministic single record: Platform: "Netflix" | Content: "Fauda" | Conflict: "Diplomatic_Row_Fauda". The AI's free-text response achieved 100% precision by accurately reproducing this exact sequence (Netflix → Fauda_Series → Diplomatic_Row_Fauda) with zero factual drift.

Coverage & Semantic Generalization: While the graph query was constrained strictly to the direct conflict path, the AI demonstrated broader semantic coverage by synthesizing additional contextual knowledge from the graph, incorporating the secondary cultural debate (Scorsese_Cinema_Debate) triggered by movie theater decline (Movie_Theater_Decline).

- **Model 2 (Perplexity):**

Precision & Determinism: The Neo4j query aggregated multi-dimensional entities into a single record: streamingPlatform: "Netflix", relatedContent: ["Squid Game"], relatedPolicies: ["EU AVMSD Policy"], and competingTraditionalOutlets: ["Kan 11"]. The AI's free-text response maintained high precision by explicitly acknowledging that direct content-to-diplomatic conflict links were absent from its schema, refusing to assert false connections.

Coverage & Semantic Generalization: While the database query required complex aggregation functions to avoid null failures, the AI flexibly summarized existing platform dynamics, explaining how soft power (StreamingSoftPowerUSImpact) and broadcast competition (Kan11Channel) were represented instead.

- **Model 3 (ChatGPT):**

Precision & Determinism: The Neo4j database returned exact RDF URI pairs confirming direct industry rivals: Netflix → DisneyPlus and Netflix → BBCOne. The AI's textual response showed perfect precision by stating that while individual content items were not linked to geopolitical conflicts, direct platform-level competition with Disney Plus and BBC One was fully verified.

Coverage & Semantic Generalization: The database returned isolated entity pairs, whereas the AI synthesized the broader graph narrative, explaining how international productions (SquidGame, MoneyHeist, Fauda) function as soft power for South Korea and cultural export mechanisms to the United States.

Competency Question 2: Geopolitical Regulation & Content Restrictions

- **Model 1 (Gemini):**

Precision & Determinism: The Neo4j database successfully traversed the complete four-node regulatory chain: Region: "Middle_East_Region" → CensorshipPolicy: "Turkish_State_Censorship" → BlockedContent: "Fauda_Series" ← AffectedPlatform: "Netflix". The AI's textual response achieved complete parity with the database, describing the exact same regulatory sequence without error.

Coverage & Semantic Generalization: Both retrieval methods demonstrated equal, comprehensive coverage due to the explicit modeling of regional censorship and content blocking in Gemini's ontology.

- **Model 2 (Perplexity):**

Precision & Determinism: The Neo4j database returned NO RECORDS RETURNED because the multi-hop query demanded a continuous path across country, policy, and impact nodes that did not form a single connected path. The AI demonstrated bounded precision by correctly explaining that specific state censorship rules blocking streaming titles were unmodeled, avoiding false positive claims.

Coverage & Semantic Generalization: Despite the zero-record return in Neo4j, the AI exhibited semantic resilience, bridging the conceptual gap by directing attention to the platform-level directive (EUAVMSDStreamingPolicy) responding to cultural homogenization (NetflixCulturalHomogenizationGlobalImpact).

- **Model 3 (ChatGPT):**

Precision & Determinism: The Neo4j query returned the formal tripartite regulatory relationship: Policy: "EuropeanContentQuota" → Platform: "Netflix" → Regulator: "EuropeanUnionMediaRegulator". The AI mirrored this result with high precision, confirming the regulatory oversight while clarifying that individual content blocking was unmodeled.

Coverage & Semantic Generalization: The graph database confirmed the policy assertion, while the AI provided conceptual disambiguation, distinguishing between mandatory local content quotas (EuropeanContentQuota requiring 30% local production) and state-level censorship.

Competency Question 3: Behavioral Shifts & Demographic Audience Impact

- **Model 1 (Gemini):**

Precision & Determinism: The Neo4j database verified both the primary impact and its secondary correlation: Platform: "Netflix" → DemographicGroup: "Youth_Cohort_18_25" → PrimaryImpact: "Binge_Sleep_Loss" → SecondaryImpact: "Teen_Isolation". The AI's response demonstrated complete precision, reconstructing the exact multi-hop behavioral pathway.

Coverage & Semantic Generalization: While the graph query retrieved only the specified impact variables, the AI expanded coverage by identifying parallel unqueried impacts (Youth_Reading_Drop) associated with the same youth demographic.

- **Model 2 (Perplexity):**

Precision & Determinism: The Neo4j query retrieved algorithmic personalization features tied to young adults (NetflixMainRecommendationAlgorithm → ViewingHabitChangeImpact → IsraeliYoungAdults18to34Group), while unmodeled fields evaluated to null. The AI showed precise boundary awareness by explaining that specific physiological impacts (such as sleep loss or isolation) were not instantiated in its graph.

Coverage & Semantic Generalization: The database output raw values, while the AI provided a holistic synthesis of audience segments, explaining how HeavyBingeWatchersClusterProfile and GlobalMiddleClassViewersGroup interact with recommendation algorithms and viewing habit changes.

- **Model 3 (ChatGPT):**

Precision & Determinism: The Neo4j query confirmed the platform-level behavioral link: Platform: "Netflix" → BehaviorImpact: "GlobalBingeWatchingGrowth". The AI's response remained strictly precise, acknowledging this verified link while explicitly stating that demographic segmentation and correlated physiological harms (sleep loss or isolation) were absent.

Coverage & Semantic Generalization: While the database was limited to a single relationship, the AI contextualized audience communities, explaining how specific target audiences like KoreanWaveCommunity (via Squid Game) and IsraeliTelevisionAudience (via Fauda) were modeled across content titles.

Evaluating the retrieval paradigms across all three models demonstrates their complementary strengths: Structured Cypher queries (Neo4j) provide deterministic precision, auditable truth, and mathematical certainty over explicit graph topologies, but are vulnerable to schema rigidity and join path mismatches; conversely, Free-Text AI retrieval delivers semantic flexibility, contextual synthesis, and grounded boundary awareness across heterogeneous schemas.

PART 4: Wikidata SPARQL Queries & Perspective Analysis

This following comparative analysis examines how global data ecosystems model and represent the cultural, geopolitical, and behavioral dimensions of streaming platforms, and evaluates whether these decentralized public repositories reflect differing institutional perspectives compared to the controlled ontological models.

4.1 Mapping Wikidata Queries to Project Competency Questions

To establish structural alignment with the Neo4j analyses, three distinct SPARQL queries were executed on the Wikidata Query Service, targeting international distribution, European regulatory frameworks, and socio-behavioral taxonomies:

1. **Query 1 (Non-US Productions Distributed by Global Platforms):** Examines international television series originating outside the United States (*Squid Game* from South Korea, *Money Heist* and *Elite* from Spain, and *Lupin* from France) that are globally distributed by Netflix (Wikidata property wdt:P750 filtered by wd:Q907311).
2. **Query 2 (European Union Media Regulations & Directives):** Retrieves formal EU legislative frameworks governing digital markets and online audiovisual media, specifically focusing on the *Audiovisual Media Services Directive (AVMSD)* and the *General Data Protection Regulation (GDPR)* (wd:Q56856283 and wd:Q1172506).
3. **Query 3 (Digital Media Behavioral & Health Taxonomies):** Maps standardized medical and behavioral taxonomies associated with modern digital media consumption, specifically retrieving entries for *binge-watching* (wd:Q15094181) and *sleep disorder* (wd:Q177190).

The full SPARQL code, direct query links, and resulting execution tables are comprehensively documented in **Appendix F**.

4.2 Comparative Perspective Analysis: Neo4j vs. Wikidata

Comparing the structured retrieval results from the Neo4j graphs with the open-world SPARQL results from Wikidata, reveals significant differences in perspective, governance, and epistemic framing across three core analytical axes:

- **Geopolitical & Cultural Distribution Perspective:**
 - *In Neo4j:* The models (*Gemini*, *Perplexity*, and *ChatGPT*) framed international content (*Squid Game*, *Fauda*) primarily through the lens of strategic corporate distribution, national soft power generation, and localized market impact (e.g., Netflix distributing content to specific regional cohorts or triggering geopolitical diplomatic tensions).
 - *In Wikidata:* Wikidata treats international series as objective bibliographic and encyclopedic entities defined by rigid administrative attributes—specifically country of origin (wdt:P495), original language (wdt:P364), and formal global distributors (wdt:P750). While Neo4j highlights the *friction and strategic intent* behind content circulation, Wikidata records the *structural reality* of transnational media flows, confirming neutral metadata such as South Korean origin for *Squid Game* and Spanish origin for *Money Heist*.
- **Regulatory & Institutional Governance Perspective:**
 - *In Neo4j:* Policies such as the EU AVMSD or content quotas were modeled as active regulatory constraints designed to shape platform behavior, protect local industries, or respond to cultural homogenization.
 - *In Wikidata:* Regulatory directives are cataloged as formal legal instruments issued by a supranational authority. Query 2 highlights formal legislative acts (*AVMSD 2010* and *GDPR*), representing governance from an institutional, compliance-oriented standpoint rather than as a reactive counter-measure to platform dominance. This demonstrates that while the closed graphs treat regulation as a dynamic variable in an industry power struggle, Wikidata maps it as part of a static legal ontology.
- **Behavioral & Health Impact Perspective:**
 - *In Neo4j:* Behavioral outcomes such as binge-watching, sleep disruption, and teen isolation were modeled as causal consequences, direct socio-behavioral impacts inflicted by platform algorithmic design on vulnerable demographic cohorts (e.g., youth cohorts aged 18–25).
 - *In Wikidata:* Concepts like *binge-watching* and *sleep disorder* appear as disconnected, standardized taxonomical nodes within a vast medical and sociological classification tree, devoid of direct causal arrows linking them to specific corporate entities like Netflix or algorithmic design choices.

4.3 Conclusion on Knowledge Representation Paradigms

The comparative analysis demonstrates that closed-world Knowledge Graphs (Neo4j) excel at narrative-driven, subjective, and causal modeling, allowing researchers to encode specific arguments regarding controversy, conflict, and societal harm. Conversely, open-world knowledge bases (Wikidata) provide an objective, consensus-driven structural backbone that validates the factual existence of entities, laws, and phenomena. Integrating both paradigms provides a robust scholarly framework, grounding the subjective media critique within verifiable global data standards.